Ramy Lazghab

AI Engineer — LLMs & Agentic AI Specialist

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Profile

M.Sc. Data Science& AI candidate at Université Paris Dauphine-PSL, specializing in Generative AI, LLMs, and MLOps. Production-ready experience building multi-agent AI systems, RAG pipelines, and temporal forecasting models on GCP and cloud infrastructure. Currently developing regulatory forecasting AI systems leveraging Anthropic SDK, LangGraph, and ensemble ML techniques. Seeking full-time AI Engineer roles to deploy scalable intelligence systems.

Technical Skills

Languages: Python (Expert), Java, C, R, JavaScript, SQL
AI & ML Frameworks: PyTorch, TensorFlow, PyTorch Lightning, Hugging Face, LangChain, scikit-learn, XGBoost, LightGBM, Transformers
Agentic AI: Anthropic SDK (Agent 2.0), LangGraph, LangSmith, prompt caching, multi-hop reasoning, autonomous tool-use
MLOps & Cloud: Docker, Git, MLflow, Vertex AI, BigQuery, Cloud Run, Streamlit, GCP, Azure (Exposure)
Data & Databases: Pandas, NumPy, Spark, PostgreSQL, Django REST, FAISS, Qdrant, Neo4j, Vector DBs
Advanced Techniques: RAG, Temporal Fusion Transformers, LoRA fine-tuning, prompt engineering, data pipelines, scientific document processing

Current Experience

AI Engineer Intern – Relay X April 2026 – Present | Paris
Regulatory Forecasting AI System

- Architected and deployed a production **multi-agent AI system** predicting EU cosmetic ingredient regulatory restrictions 12–24 months in advance, covering raw data ingestion to end-user intelligence dashboard.
- Implemented **Anthropic SDK Agent 2.0** orchestrator with autonomous tool-use loops, multi-hop reasoning across 6 live regulatory data sources (PubMed, PubChem, EUR-Lex, CosIng), and **prompt caching** achieving 90% LLM cost reduction per session.
- Built **LangGraph parallel data pipeline** with real-time fetching, full **LangSmith observability** tracing every node execution and system performance.
- Engineered **RAG extraction pipeline** over 511 SCCS scientific opinion PDFs—semantic chunking (LangChain), Qdrant vector store (50,816 chunks), OpenAI embeddings, and Claude-powered synthesis—converting unstructured regulatory documents into structured ML-ready dataset.
- Delivered **17-step preprocessing pipeline** and **XGBoost + LightGBM ensemble** risk scorer validated via strict temporal cross-validation (train 2019, test 2020–2025), achieving **93.7% accuracy with zero data leakage**.
- Designed **unified Temporal-Semantic-Causal Reasoning (TSCR) architecture**—SciBERT + ChemBERTa + Temporal Fusion Transformer with LoRA fine-tuning—for multimodal fusion of regulatory text, chemical properties, and time-series signals.
- Architected **dynamic knowledge graph layer** (Neo4j + GNN with temporal causal attention) autonomously discovering cross-source causal chains between regulatory events, chemical structures, and political contexts.
- Built **8-tab Streamlit intelligence dashboard** covering live literature monitoring, EUR-Lex amendment history, ingredient risk scoring, and AI-generated regulatory narratives.

Previous Experience

Software Developer Intern – Infinity Management 02/2024–06/2024

- Built **NLP-driven adaptive learning platform** using BERT-based sentiment analysis for personalized user engagement and retention.
- Containerized full **React + Django REST** system with Docker; implemented distributed logging and monitoring for scalable production deployment.

Web Developer Intern – Centre National d'Informatique 06/2023

- Developed secure client-registration web application (PHP/HTML/CSS) optimizing internal data workflows and user onboarding efficiency.

AI & Data Projects

- **TelecomPlus Multi-Agent Support System** – Multi-agent architecture combining RAG over PDF FAQs, SQL data access, and LangChain orchestration. Integrated LLM-as-a-judge evaluation and Langfuse/MLflow monitoring. *Tools: LangChain, OpenAI API, Pandas, Faiss, Streamlit*
- **RAGenius** – LLM-powered research assistant for CSV/PDF analysis; optimized retrieval achieving **92% precision** on complex QA tasks. *Tools: LangChain, Hugging Face, Faiss, Python*
- **Movie Recommendation System on GCP** – End-to-end personalized recommender with BigQuery + Vertex AI + Cloud Run; deployed adaptive REST API. *Tools: BigQuery, Vertex AI, Cloud Run, Python*

- **Alzheimer's Diagnosis Classifier (Kaggle)** – Handwriting-feature classifier using ensemble models (XGBoost, Deep Learning); **AUC = 0.96**; applied SHAP for clinical interpretability. *Tools: XGBoost, PyTorch, SHAP, Python*
- **MoodSync (Hack for Good Winner)** – Real-time therapeutic AI assistant with speech-emotion recognition and IoT ambient lighting. *Tools: Deep Learning, OpenAI, IoT APIs*
- **SportiQ** – AI-powered sports-pose analysis using OpenCV & MediaPipe; NLP-based feedback improved correction latency by 40%. *Tools: OpenCV, MediaPipe, NLP, Python*

Education

Université Paris Dauphine-PSL 09/2024–09/2026 (Expected)
M.Sc. Data Science& Artificial Intelligence
Key Courses: Generative AI, Deep Learning, NLP, Reinforcement Learning, Big Data Systems, Graph Neural Networks

Faculty of Sciences of Tunis 09/2021–06/2024
B.Sc. Computer Engineering

Certifications & Achievements

- **Certified – LangChain Academy**
- **Certified – Google Cloud: Create ML Models With BigQuery ML (Intermediate)**
- **Winner – Hack for Good:** Designed AI platform *MoodSync* for well-being analysis and monitoring
- **Participant – Hack for Smart Insurance (EY):** Developed *InsurAI*, AI-based solution for automating insurance claim reports
- **IEEE Xtreme 15.0 & 16.0 Competitor:** Global programming competition participant
- **Active Community Member:** IEEE, ITLAB, AIESEC — organized workshops, led hackathons, mentored junior engineers

Languages

English (Fluent – IELTS) | **French** (Professional)